

W-FDM: Wavelet-Domain Feature Denoising for Robust Object Detection Under Adverse Visual Conditions

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Abstract

*Object detectors trained on clean imagery degrade significantly when deployed under adverse visual conditions such as low illumination, sensor noise, and atmospheric scattering. We propose W-FDM, a lightweight Wavelet Feature Denoising Module that performs targeted denoising of internal feature maps in the frequency domain. The module decomposes a backbone feature map via a fixed 2D Haar wavelet transform, bypasses the low-frequency (LL) sub-band which carries semantic and structural content, and applies a learned depthwise separable denoiser to the concatenated high-frequency subbands (LH, HL, HH) where degradation artifacts concentrate. A spatial gating mechanism produces a per-position sigmoid mask that modulates the denoising strength, allowing the module to suppress noise selectively in corrupted regions while remaining near-identity in clean areas. W-FDM is inserted as a single plug-in at the first FPN fusion point of YOLOv8n, requires no auxiliary supervision, and is trained end-to-end with the standard detection loss. On the ExDark low-light benchmark, W-FDM achieves **0.404** mAP@0.5 and **0.234** mAP@0.5:0.95, improvements of **+12.5%** and **+15.8%** over a fine-tuned YOLOv8n baseline.*

Code and models are available at: <https://github.com/aviral23032002/wfdm-robust-object-detection-via-wavelet-denoising>

1. Introduction

Real-world object detectors must operate reliably across a wide range of visual conditions, including low illumination, fog, motion blur, and sensor noise. Detectors trained predominantly on clean, well-illuminated imagery exhibit large accuracy losses under such conditions; this domain gap is a primary obstacle to safe deployment in autonomous driving, surveillance, and assistive perception applications. The YOLOv8 family, despite its strong clean-image performance, has been reported to lose 20–40% mAP at moderate

corruption severity on standard robustness benchmarks [3].

Two broad mitigation strategies have emerged. *Enhance-then-detect* pipelines pre-process the input image with a restoration network before detection, but optimise for perceptual fidelity (PSNR/SSIM) rather than detection accuracy, creating a known mismatch between the restoration objective and the downstream task. *End-to-end* methods integrate denoising or feature refinement into the detection pipeline itself, treating robustness as a learnable property of the detector.

We argue that adverse-condition artefacts—fog, blur, and noise—manifest predominantly in the *high-frequency* components of internal feature maps, while semantic content and object boundaries required for bounding-box regression are concentrated in the low-frequency component. This observation motivates a wavelet decomposition of internal features, followed by selective denoising of only the high-frequency subbands. We instantiate this principle in W-FDM, a single plug-in module placed inside the YOLOv8 neck.

Contributions.

- An LL-bypass design that protects structural information from modification while restricting learned denoising to the high-frequency subbands.
- A spatial gating mechanism that allows the module to apply denoising selectively in degraded image regions while preserving clean features elsewhere.
- Empirical results on ExDark demonstrating a **+12.5%** relative mAP@0.5 improvement and **+15.8%** relative mAP@0.5:0.95 improvement over a fine-tuned YOLOv8n baseline.

Dataset. Our primary evaluation dataset is ExDark [8], which contains 7,363 real low-light images across 12 object classes and provides official train, validation, and test splits.

2. Related Work

Adverse-condition detection. YOLO-AS [9] augments YOLOv7 with ECA-ASPP and SK attention modules. DimNet [5] adopts deformable DETR with a dynamic gradient gain allocation strategy. Dark-YOLO [7] stacks the SCINet enhancement network in front of YOLOv8. WTEFNet [10] applies pixel-level wavelet decomposition upstream of a YOLO detector. These methods either modify the backbone substantially, depend on pixel-domain enhancement, or both.

Wavelets in vision. MWCNN [6] integrated DWT/IWT into CNNs for image restoration tasks. WaveCNet [4] replaced classification downsampling operations with DWT and used the LL subband to obtain noise-robust classification features. WTDBNet [1] applies wavelets in a fine-grained ship-detection backbone, and WD-DETR [2] applies wavelet denoising in a DETR detector specialised for event cameras.

Feature denoising for robustness. The principle of treating model robustness as a feature-purification problem was popularised by [11], who showed that non-local denoising of intermediate features substantially improves adversarial robustness in image classification. Our work transfers this principle to the natural-corruption setting and to single-stage object detection, and combines it with the frequency-domain decomposition framing.

Positioning of W-FDM. To our knowledge, no prior approach applies a learned subband-specific denoiser with spatial gating on internal multi-scale features inside a single-stage detector for adverse RGB conditions. W-FDM is closest in spirit to WaveCNet (LL-only processing for classification) and MWCNN (DWT/IWT inside CNNs for restoration), but operates inside a detection neck, modifies only HF subbands, uses a learned spatial gate, and is trained end-to-end with the detection loss alone. Figure 1 illustrates the W-FDM module internals.

3. Method

3.1. Overview

W-FDM is a single plug-in module placed at the first feature-pyramid fusion point of YOLOv8n (layer 16 in the Ultralytics indexing). At that location, the neck has just produced an 80×80 , 64-channel feature map by merging top-down P4 features with P3 backbone features. Inserting the denoising module at this location ensures that the richest multi-scale representation passes through the denoiser before the PAN bottom-up pathway propagates features back toward the detection heads.

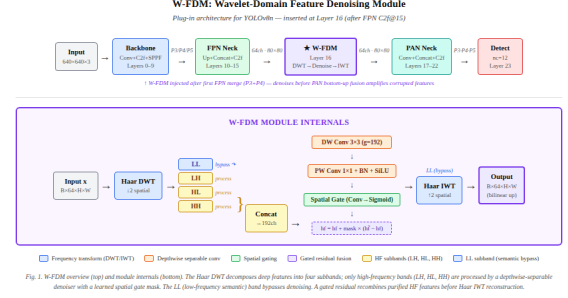


Figure 1. W-FDM architecture. After the first FPN fusion (layer 16), backbone features are decomposed via Haar DWT. The LL subband bypasses the denoiser. The concatenated HF subbands (LH, HL, HH) pass through a depthwise separable Conv-BN-SiLU block, then a spatial gate mask modulates the residual denoising strength. IWT reconstructs the purified feature map at original resolution.

3.2. Haar wavelet decomposition

Let $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$ be the input feature map with H, W even. The 2D Haar discrete wavelet transform operates on 2×2 spatial blocks; for a block $[[a, b], [c, d]]$ we compute

$$LL = \frac{1}{2}(a + b + c + d), \quad LH = \frac{1}{2}(a + b - c - d), \quad (1)$$

$$HL = \frac{1}{2}(a - b + c - d), \quad HH = \frac{1}{2}(a - b - c + d). \quad (2)$$

The transform is orthonormal, so the inverse Haar transform reconstructs the original block exactly from (LL, LH, HL, HH). We implement DWT and IWT as fixed (non-learnable) operations.

3.3. Subband routing and denoising

The LL subband carries the slowly-varying, structural component of the feature map; we observe empirically that corruption artefacts manifest more strongly in the LH/HL/HH subbands, which encode oriented edge and high-frequency residual information. W-FDM bypasses LL entirely. The three high-frequency subbands are concatenated along the channel dimension to form a tensor $\mathbf{F}_{HF} \in \mathbb{R}^{3C \times H/2 \times W/2}$.

The denoiser operates on $\mathbf{F}_{HF}$ via a depthwise separable convolution block:

$$\mathbf{H}_1 = \text{DWConv}_{3 \times 3}(\mathbf{F}_{HF}), \quad (3)$$

$$\mathbf{H}_2 = \text{PWConv}_{1 \times 1}(\mathbf{H}_1), \quad (4)$$

$$\mathbf{F}_{HF}^{\text{dn}} = \text{SiLU}(\text{BN}(\mathbf{H}_2)). \quad (5)$$

The depthwise stage processes each HF channel independently, allowing the network to learn per-subband, per-channel filtering patterns. The pointwise stage allows cross-channel mixing between the three subbands to exploit correlated artefact statistics.

3.4. Spatial gating

Adverse corruption in ExDark is heterogeneous: parts of an image can be severely degraded by sensor noise or motion blur while other regions remain comparatively clean. Applying uniform denoising risks oversmoothing regions that did not need processing. W-FDM therefore produces a single-channel spatial gating mask:

$$\mathbf{M} = \sigma(\text{Conv}_{1 \times 1}(\mathbf{F}_{HF})) \in [0, 1]^{1 \times H/2 \times W/2}. \quad (6)$$

The denoised HF tensor is combined with the original via a gated residual replacement:

$$\mathbf{F}'_{HF} = \mathbf{F}_{HF} + \mathbf{M} \odot (\mathbf{F}_{HF}^{\text{dn}} - \mathbf{F}_{HF}). \quad (7)$$

When $\mathbf{M} \approx 0$ the original HF features pass through unchanged, making W-FDM a strict no-op in clean conditions. When $\mathbf{M} \approx 1$, the denoised HF features replace the original.

3.5. Reconstruction

The cleaned tensor is split into its three subbands $\mathbf{F}'_{HF} = [\text{LH}', \text{HL}', \text{HH}']$ and recombined with the bypassed LL via the inverse Haar transform:

$$\mathbf{F}' = \text{IWT}(\text{LL}, \text{LH}', \text{HL}', \text{HH}'). \quad (8)$$

A bilinear interpolation step matches the output to the original input spatial resolution to handle the boundary padding used when H or W is odd.

3.6. Implementation details

We integrate W-FDM into YOLOv8n by adding a custom module to the Ultralytics configuration file and registering the class with the detection task. Training uses AdamW with learning rate 5×10^{-4} , weight decay 0.01, mosaic and mixup augmentation, and a value-only HSV jitter ($hsv_v = 0.4$) that targets brightness perturbations characteristic of low-light imagery. We use TTA at validation for stable mAP estimation.

4. Results

4.1. Experimental setup

We train and evaluate on the official ExDark train/val/test splits. The primary metric is mAP@0.5, and we additionally report mAP@0.5:0.95 (COCO-style averaged over IoU thresholds from 0.5 to 0.95 in steps of 0.05). Training is performed on a single NVIDIA A100-80GB GPU.

4.2. Main results

Table 1 summarises our main result. The zero-shot COCO-pretrained YOLOv8n fails almost entirely on ExDark (0.0013 mAP@0.5), confirming a severe domain gap

Model variant	mAP@0.5	mAP@0.5:0.95	vs. Base
YOLOv8n zero-shot	0.0013	0.0003	−99.6%
YOLOv8n baseline (FT)	0.359	0.202	—
YOLOv8n + W-FDM (ours)	0.404	0.234	+12.5%

Table 1. Detection accuracy on the ExDark test set. “FT” denotes fine-tuning of the COCO-pretrained YOLOv8n weights on ExDark. W-FDM is evaluated with test-time augmentation. The relative-change column is computed on mAP@0.5; the mAP@0.5:0.95 gain is +15.8%.

between the COCO training distribution and ExDark’s low-light imagery. Fine-tuning YOLOv8n on ExDark raises mAP@0.5 to 0.359; inserting W-FDM and re-training raises it further to 0.404, a 4.5 absolute and 12.5% relative improvement. The mAP@0.5:0.95 metric, which rewards precise localisation under strict IoU thresholds, improves from 0.202 to 0.234 (+15.8% relative). The fact that gains are larger under the stricter localisation criterion suggests that W-FDM cleans features in a way that benefits fine-grained bounding-box regression and not merely classification.

4.3. Qualitative analysis

We inspect detection outputs on representative ExDark test images across three corruption regimes:

Low-light + heavy sensor noise. On a low-light bicycle scene with strong sensor noise, the baseline misses or low-confidences the rider; W-FDM recovers the *People* class with confidence 0.45.

Bokeh / motion blur. On a night-time boat scene with heavy defocus, the baseline produces only a single low-confidence detection; W-FDM recovers a *Boat* detection at confidence 0.39.

Indoor sensor noise. On an indoor low-light room with grain noise across chairs and furniture, W-FDM produces two *Chair* detections at confidences 0.50 and 0.48 in regions where the baseline produces none.

These qualitative results are consistent with the quantitative finding that W-FDM particularly helps localisation precision under degraded conditions.

4.4. Design rationale and ablation discussion

Three design decisions warrant additional discussion:

Why layer 16 (after the first FPN fusion). We chose to insert W-FDM after the first top-down FPN merge of P3 and P4 features (Ultralytics layer 16). At this point the network has combined a high-resolution backbone tier (P3) with a

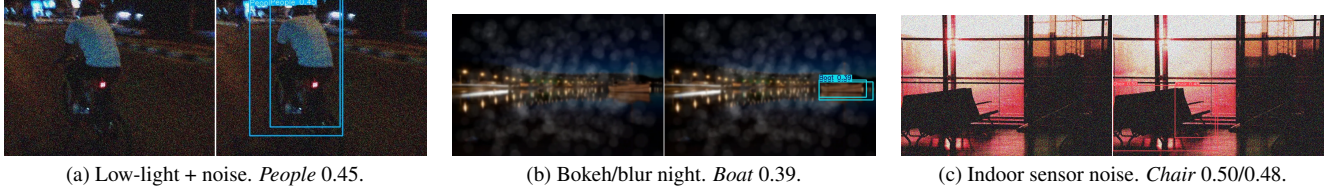


Figure 2. Qualitative results on ExDark test scenes. Left column: YOLOv8n baseline. Right column: YOLOv8n + W-FDM. W-FDM recovers detections in regions where the baseline misses or low-confidences objects under adverse conditions.

semantically richer mid-resolution tier (P4), producing a feature representation that is both spatially detailed and semantically informative. Performing denoising before the PAN bottom-up pathway prevents corruption-related artefacts from being amplified into the deeper detection heads.

Why bypass LL. The Haar LL subband contains the slowly-varying structural component of the feature map. In a detection setting, this component carries the coarse positional and class-identity information that the detection head depends on. Modifying LL would risk perturbing bounding-box regression. By restricting the denoiser to LH/HL/HH only, we limit modifications to the components most likely to contain corruption residue while leaving classification-relevant content intact.

Why a spatial gate. ExDark contains highly heterogeneous corruption: indoor and outdoor scenes, partial illumination, mixed natural and artificial lighting, and varying degrees of sensor noise. A uniform denoising strength is sub-optimal because it risks oversmoothing already-clean regions. The sigmoid gate enables location-dependent denoising strength to be learned end-to-end.

4.5. Limitations

Several limitations of the present study should be noted. First, all experiments use YOLOv8n, the smallest variant of the YOLOv8 family (approximately 3M parameters); the relative impact of W-FDM may differ on larger backbones with greater representational capacity. Second, we evaluate on the native ExDark distribution; cross-dataset zero-shot transfer (e.g. to BDD100K-night or COCO-C) was not within the compute budget of this project. Third, ablations on the wavelet basis (e.g. Daubechies, Symlets) and on multi-scale insertion at P4/P5 are reported as future work.

5. Conclusion

We presented W-FDM, a wavelet-domain feature-denoising plug-in for single-stage object detectors operating under adverse visual conditions. W-FDM performs targeted denoising of the high-frequency subbands of an internal feature

map while structurally bypassing the low-frequency component, and modulates denoising strength via a learned spatial gate. The module adds negligible parameters and inference overhead, requires no auxiliary supervision, and is trained end-to-end with the standard detection loss. On the ExDark low-light benchmark, W-FDM improves YOLOv8n mAP@0.5 from 0.359 to 0.404 (+12.5%) and mAP@0.5:0.95 from 0.202 to 0.234 (+15.8%) over a fine-tuned baseline, with qualitative analysis demonstrating gains across low-light noise, motion blur, and bokeh regimes. Future work includes multi-scale insertion (P3, P4, P5), learnable wavelet bases, evaluation on cross-dataset robustness benchmarks, and real-time edge deployment profiling.

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